# Yuhui Ding | Curriculum Vitae

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skeletondyh

## Education

**The Hong Kong University of Science and Technology** **Hong Kong**
*M.Phil., Department of Computer Science and Engineering* *Sept. 2019 – Nov. 2021*
Supervisor: Prof. Tong Zhang
Thesis Title: Neural Architecture Search for Heterogeneous Graph Neural Networks

**Tsinghua University** **Beijing**
*B.Eng., Department of Computer Science and Technology* *Aug. 2015 – July 2019*

## Publications

1. **Yuhui Ding**, Quanming Yao, Huan Zhao, Tong Zhang. DiffMG: Differentiable Meta Graph Search for Heterogeneous Graph Neural Networks. *KDD*, 2021.
2. Zhengxiao Du, Jie Tang, **Yuhui Ding**. POLAR++: Active One-shot Personalized Article Recommendation. *IEEE Transactions on Knowledge and Data Engineering* (*TKDE*), 2019.
3. Changjun Fan, Li Zeng, **Yuhui Ding**, Muhao Chen, Yizhou Sun, Zhong Liu. Learning to Identify High Betweenness Centrality Nodes from Scratch: A Novel Graph Neural Network Approach. *CIKM*, 2019.
4. **Yuhui Ding**, Shanshan Li, Jinghan Liu, Xiaojun Wu. DCLab: A Web-based System for Digital Logic Experiment Teaching. *IEEE Frontiers in Education Conference* (*FIE*), 2018.
5. Zhengxiao Du, Jie Tang, **Yuhui Ding**. POLAR: Attention-based CNN for One-shot Personalized Article Recommendation. *ECML-PKDD*, 2018.

## Research Experience

**The Hong Kong University of Science and Technology**
*Research assistant, supervised by Prof. Tong Zhang* *Sept. 2019 – Present*

- Proposed a neural architecture search method for GNNs to automatically utilize task-related semantic information encoded in heterogeneous information networks. The proposed method eliminated the need for handcrafted rules, required little search time, and achieved better performance than previous heterogeneous GNNs. One first-author paper has been accepted by KDD 2021 Research Track.
- Proposed a stochastic bi-level optimization algorithm based on implicit differentiation and applied it to tuning $L_2$ regularization.

**Tsinghua University**
*Research assistant, supervised by Prof. Jie Tang* *Nov. 2016 – Nov. 2018*
Contributed to personalized article recommendation via one-shot learning and active learning. Papers have been published in ECML-PKDD 2018 and TKDE, and our model has been launched on AMiner.

**University of California, Los Angeles**
*Research assistant, supervised by Prof. Yizhou Sun* *July 2018 – Nov. 2018*
Contributed to efficient approximation of betweenness centrality with graph neural networks. Our work has been published in CIKM 2019.

## Work Experience

**Microsoft Research Asia**
*Intern* *Summer 2021*
Contributed to the open source AutoML toolkit NNI.

**4Paradigm Inc.**
*Part-time intern* *Oct. 2019 – Feb. 2021*
Worked on NAS for GNNs.

## Selected Course Projects

**DCLab**
*Tsinghua University*
DCLab is a software system for online digital logic experiment teaching. It supports circuit diagrams, VHDL programming, simulation, etc. I led a group of four to develop it using JavaScript. Functions of DCLab have been used in the digital logic course, and a paper describing the system has been published in proceedings of FIE 2018 (a top conference focusing on innovations related to engineering education). On my website there is a demo showing a counter built with 74LS chips.

## Awards

**Postgraduate Studentship**
*The Hong Kong University of Science and Technology* *2019 – 2021*

**Outstanding Graduate**
*Department of Computer Science and Technology, Tsinghua University* *2019*

**Innovation Excellence**
*Tsinghua University* *2017*
Only 4 in the whole class of more than 150 students.

**Third Prize**
*35th "Challenge Cup" (national extracurricular competition of science and technology)* *2017*
A prototype system that could automatically sort waste based on a CNN and a Raspberry Pi.

**Academic Excellence**
*Tsinghua University* *2016*
Top 10% of more than 140 students.

## Service

Reviewer for NeurIPS, ICLR

## Skills

**Programming Languages**: Python, C/C++, MATLAB, Java, JavaScript

**Deep Learning**: TensorFlow, PyTorch